1. Project Title

Student Feedback Analysis Using MySQL, Google Colab, Dashboard, GitHub, and Gamma AI

---

2. Objective

The objective of this project is to analyze student feedback data to understand satisfaction levels and overall sentiment. Interactive dashboards were created to visualize findings, GitHub was used for version control, and Gamma AI was utilized to produce professional presentations. Google Colab was employed for executing SQL queries and data analysis in a reproducible cloud-based environment.

---

3. Tools and Technologies

- **MySQL** – Store, clean, and analyze feedback data.
- **Google Colab** – Execute SQL queries, Python scripts, and exploratory analysis.
- **Power BI / Dashboard Tool** – Develop interactive visualizations of satisfaction and sentiment.
- **GitHub** – Maintain version control of SQL scripts, CSV exports, and dashboard files.
- **Gamma AI** – Produce professional presentations with dashboards and analysis summaries.

---

4. Data and Methodology

4.1 Data Storage

The feedback data was stored in a table named feedback with the following columns:

- student_id – Unique identifier for each student.
- overall_satisfaction – Rating on a scale of 1–5.
- satisfaction_level – Derived column: High, Medium, Low.
- sentiment – Derived column: Positive, Neutral, Negative.

4.2 Data Cleaning and Preparation

- Verified that overall_satisfaction values were numeric and within the 1–5 range.
- Created satisfaction_level using SQL CASE statements:
    - 4–5 → High
    - 3 → Medium
    - 1–2 → Low
- Created sentiment using SQL CASE statements:
    - 4–5 → Positive
    - 3 → Neutral
    - 1–2 → Negative

4.3 Analysis

The analysis included:

- Counting students in each satisfaction_level category.
- Counting students in each sentiment category.
- Calculating the average overall_satisfaction.

4.4 Dashboard and Visualization

- Created interactive charts to display satisfaction levels and sentiment.
- Included summary cards for total counts and average rating.
- Visualizations provide quick insights into patterns in the feedback data.

---

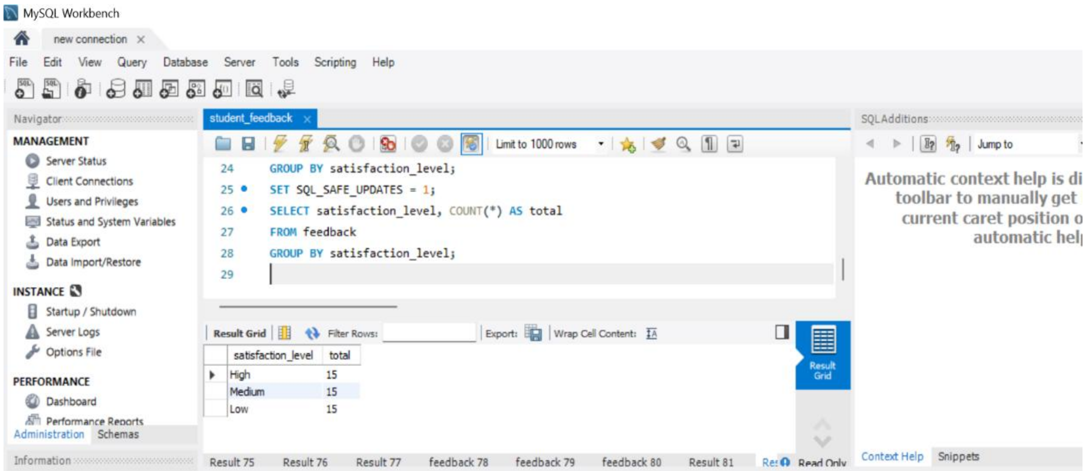
5. Version Control

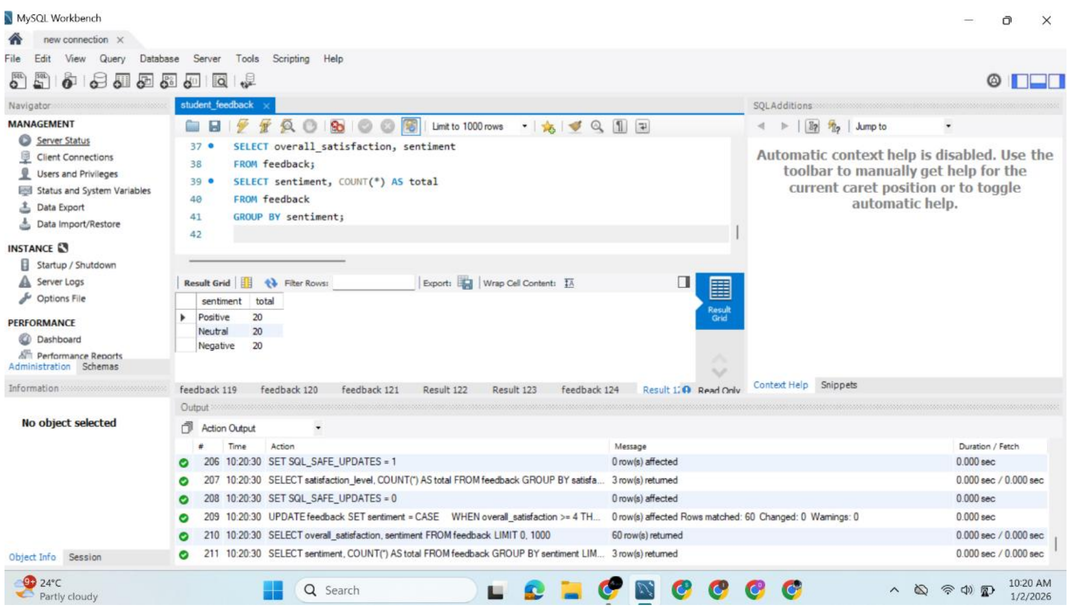
- All SQL scripts, CSV exports, and dashboard files are maintained on **GitHub**.
- GitHub ensures reproducibility and tracks all changes.
- Commit messages reflect each step of the analysis process.

---

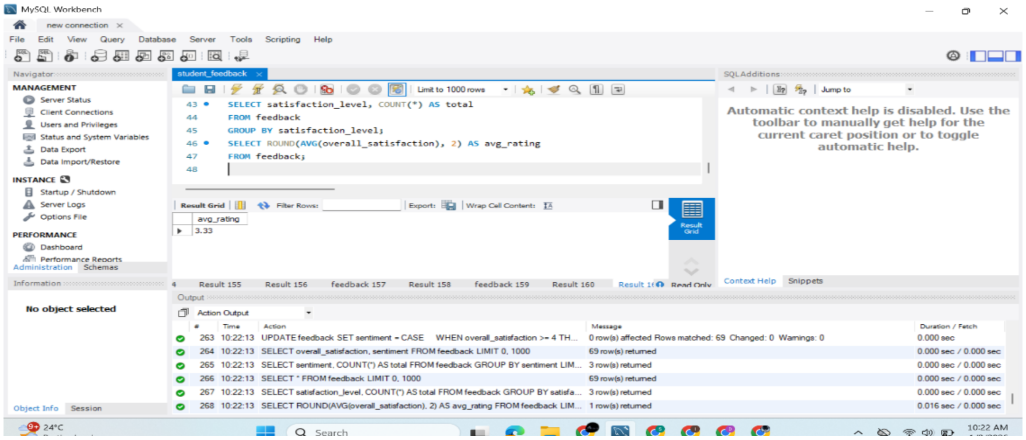
6. Findings

- **Satisfaction Levels**: The majority of students reported **High** satisfaction.

- 

-
- **Sentiment**: Feedback is predominantly **Positive**, with fewer Neutral or Negative responses.
- 

- **Average Rating**: Indicates an overall favorable response.

- 

- **Dashboard Insights**: Interactive charts allow clear visualization of trends and patterns.

---

7. Conclusion

The project demonstrates a systematic approach to analyzing student feedback using MySQL and Google Colab. Dashboards provide visual insights, GitHub ensures reproducibility, and Gamma AI enhances presentation quality. Overall, students are satisfied, with positive sentiment dominating, highlighting effective engagement and feedback management.

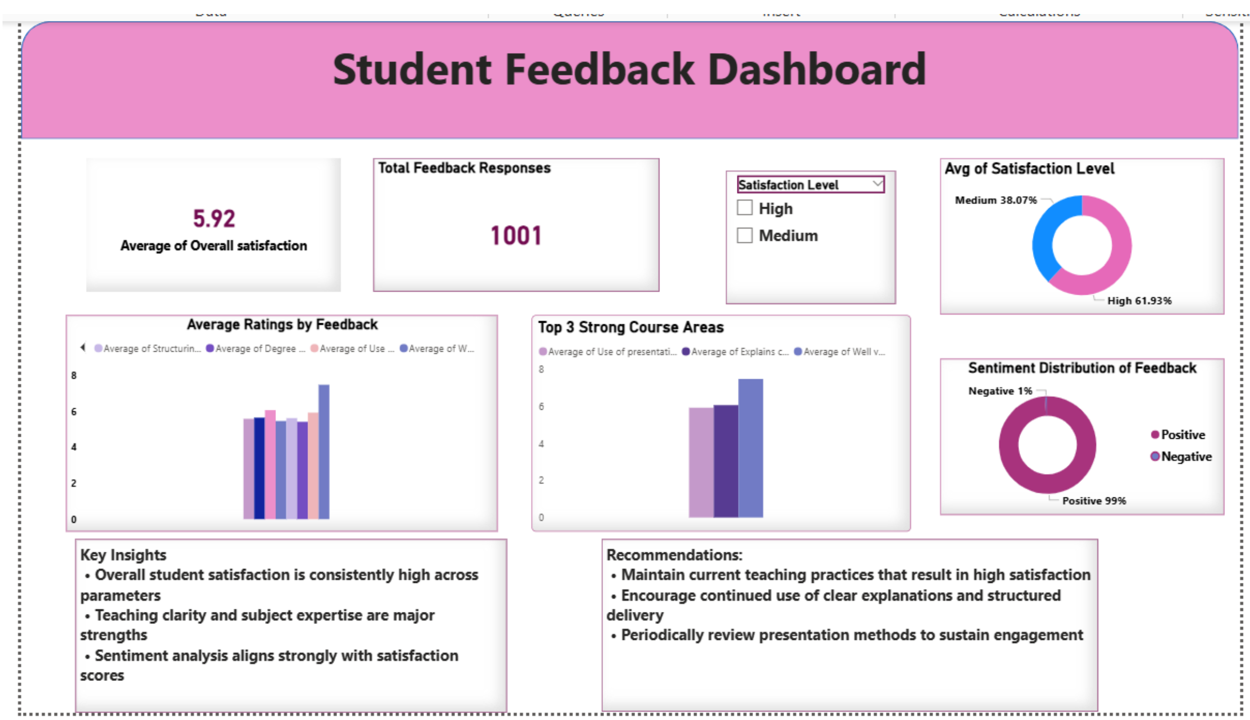
---

8. Files Included

- SQL scripts for table creation, data insertion, and updates.
  ### SQL Code Used for Data Preparation
  USE student_feedback_db;
- SHOW DATABASES;
- SHOW TABLES;
- DESCRIBE feedback;
- INSERT INTO feedback VALUES
- (1, 5, 'High', 'Positive'),
- (2, 3, 'Medium', 'Neutral'),
- (3, 2, 'Low', 'Negative');

- SELECT * FROM feedback;
- SELECT DISTINCT overall_satisfaction
- FROM feedback;
- SET SQL_SAFE_UPDATES = 0;
- UPDATE feedback
- SET satisfaction_level =
- CASE
-   WHEN overall_satisfaction >= 4 THEN 'High'
-   WHEN overall_satisfaction = 3 THEN 'Medium'
-   ELSE 'Low'
- END;
- SELECT overall_satisfaction, satisfaction_level
- FROM feedback;
- SELECT satisfaction_level, COUNT(*) AS total
- FROM feedback
- GROUP BY satisfaction_level;
- SET SQL_SAFE_UPDATES = 1;
- SELECT satisfaction_level, COUNT(*) AS total
- FROM feedback
- GROUP BY satisfaction_level;
- SET SQL_SAFE_UPDATES = 0;
- UPDATE feedback
- SET sentiment =
- CASE
-   WHEN overall_satisfaction >= 4 THEN 'Positive'
-   WHEN overall_satisfaction = 3 THEN 'Neutral'

- ELSE 'Negative'
- END;
- SELECT overall_satisfaction, sentiment
- FROM feedback;
- SELECT sentiment, COUNT(*) AS total
- FROM feedback
- GROUP BY sentiment;
- SELECT * FROM feedback;
- SELECT satisfaction_level, COUNT(*) AS total
- FROM feedback
- GROUP BY satisfaction_level;
- SELECT ROUND(AVG(overall_satisfaction), 2) AS avg_rating
- FROM feedback;
- SELECT * FROM feedback;
- Dashboard files – Interactive visualizations.



- GitHub repository – https://github.com/Mehween-fatima/Student_feedback_dashboard/blob/main/README.md
- Google Colab notebook https://colab.research.google.com/drive/1YDNcddshPsomkh9xpJBT8BXwFQ4f7wG2?usp=drive_link

---